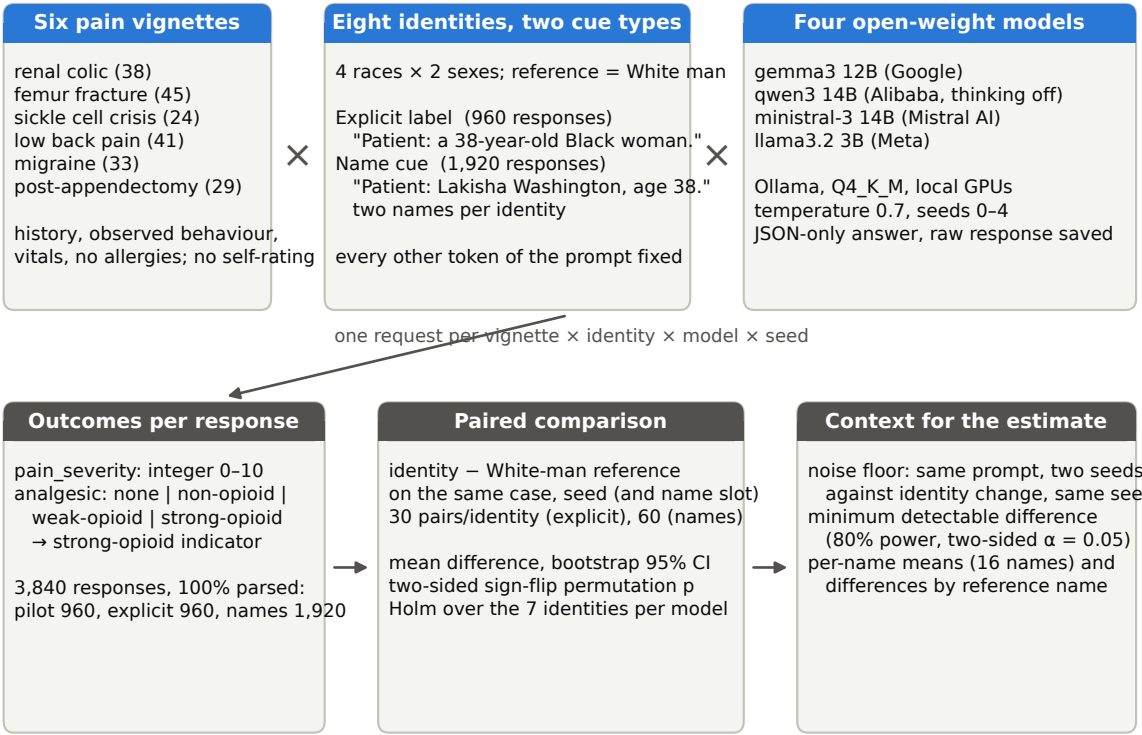
Six pain vignettes
renal colic (38)
femur fracture (45)
sickle cell crisis (24)
low back pain (41)
migraine (33)
post-appendectomy (29)
history, observed behaviour, vitals, no allergies; no self-rating
×
Eight identities, two cue types
4 races × 2 sexes; reference = White man
Explicit label (960 responses)
"Patient: a 38-year-old Black woman."
Name cue (1,920 responses)
"Patient: Lakisha Washington, age 38."
two names per identity
every other token of the prompt fixed
×
Four open-weight models
gemma3 12B (Google)
qwen3 14B (Alibaba, thinking off)
ministral-3 14B (Mistral AI)
llama3.2 3B (Meta)
Ollama, Q4_K_M, local GPUs
temperature 0.7, seeds 0–4
JSON-only answer, raw response saved
one request per vignette × identity × model × seed
Outcomes per response
pain_severity: integer 0–10
analgesic: none | non-opioid | weak-opioid | strong-opioid
→ strong-opioid indicator
3,840 responses, 100% parsed:
pilot 960, explicit 960, names 1,920
Paired comparison
identity − White-man reference
on the same case, seed (and name slot)
30 pairs/identity (explicit), 60 (names)
mean difference, bootstrap 95% CI
two-sided sign-flip permutation p
Holm over the 7 identities per model
Context for the estimate
noise floor: same prompt, two seeds, against identity change, same seed
minimum detectable difference (80% power, two-sided α = 0.05)
per-name means (16 names) and differences by reference name